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6.2.1 Boyle's Law: Pressure-Volume Relationship

Robert Boyle performed a series of experiments to study the relation between the pressure and volume of gases. The schematic of the apparatus used by him is shown in figure 6.1.

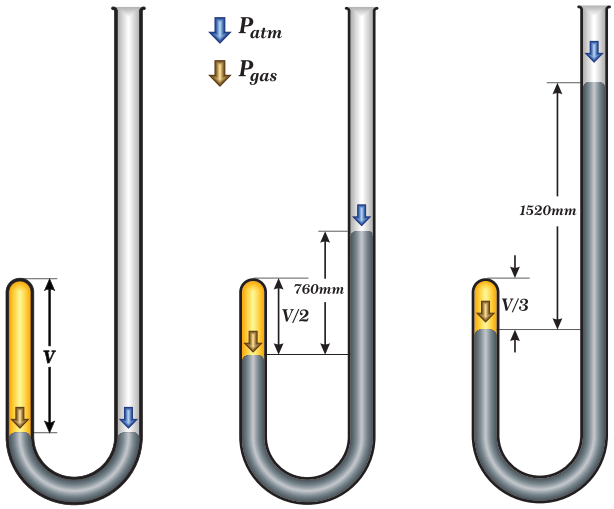


Figure. 6.1 Boyle's law experiment

Mercury was added through the open end of the apparatus such that the mercury level on both ends are equal as shown in the figure 6.1(a). Add more amount of mercury until the volume of the trapped air is reduced to half of its original volume as shown in figure 6.1(b). The pressure exerted on the gas by the addition of excess mercury is given by the difference in mercury levels of the tube. Initially the pressure exerted by the gas is equal to 1 atm as the difference in height of the mercury levels is zero. When the volume is reduced to half, the difference in mercury levels increases to 760 mm. Now the pressure exerted by the gas is equal to 2 atm. It led him to conclude that at a given temperature the volume occupied by a fixed mass of a gas is inversely proportional to its pressure.

Mathematically, the Boyle's law can be written as

$$V \propto \frac{1}{P} \quad \text{-------} \quad (6.1)$$

(T and n are fixed, T-temperature, n- number of moles)

$$V = k \times \frac{1}{P} \quad \text{-------} \quad (6.2)$$

k – proportionality constant

When we rearrange equation 6.2.

PV = k at constant temperature and mass

Boyle's law is applicable to all gases regardless of their chemical identity (provided the pressure is low). Therefore, for a given mass of a gas under two different sets of conditions at constant temperature we can write

$$P_1V_1 = P_2V_2 = k \quad \text{------} \quad (6.3)$$

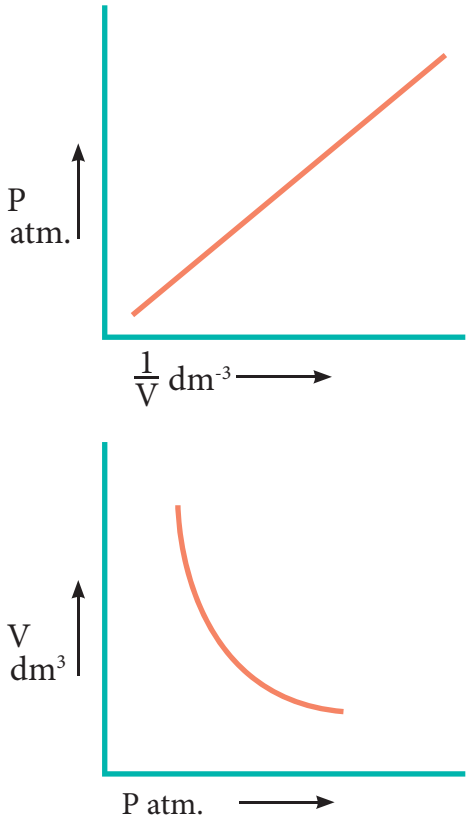


Figure. 6.2 Graphical representation of Boyle's law